1. What does my system do?

Your system, StegoGuard, detects whether images attached to emails may contain hidden information using AI.

- It works with the Thunderbird email client.
- It checks image attachments in an email.
- It uses a trained AI model to classify each image as Clean or Stego.
- It generates a Grad-CAM heatmap to show which image regions influenced the AI's prediction.
- It displays the results on a dashboard so the user can review them.

2. What is the input?

Your system receives:

- An email: An email opened in Thunderbird.
- Image attachments: Images attached to that email, such as PNG or JPEG files.
- Image data: The actual pixel information from each image, which is processed by the AI model.

The system processes each supported image attachment separately.

3. What processing happens?

Step 1: Receive the email

The user opens an email in Thunderbird.

Step 2: Find image attachments

The extension identifies supported images in the email.

Step 3: Analyse each image

The Flask backend sends the image through the trained AI model to predict whether it is clean or stego.

Step 4: Explain the prediction

Grad-CAM generates a heatmap showing regions that influenced the model's decision.

Step 5: Display the results

The dashboard displays the results for each image attachment.

4. What is the output?

For each analysed image, your system provides:

Output	Meaning
Prediction	Whether the AI classifies the image as Clean or Stego.
Clean score	The model's score for the clean class.
Stego score	The model's score for the stego class.
Grad-CAM heatmap	A visual explanation of image regions that influenced the prediction.
Image information	The attachment's filename and image information used by the dashboard.

If an email contains multiple supported images, the system can display a separate prediction, scores and heatmap for each image.